Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu_D \neq 0$	$\mu_D > 0$	$\mu_D < 0$
Rejection Rule	$\lvert t_0 \rvert \geq t_{n-1, \frac{\alpha}{2}}$	$t_0 > t_{n-1, \alpha}$	$t_0 < -t_{n-1, \alpha}$

Example 2.5

A company gave an intensive training to its salesmen to increase the sales. A random sample of 10 salesmen was selected and the value (in lakhs of Rupees) of their sales per month, made before and after the training is recorded in the following table. Test whether there is any increase in mean sales at 5% level of significance.

Salesman	1	2	3	4	5	6	7	8	9	10
Before	15	22	6	17	12	20	18	14	10	16
After	17	23	16	20	14	21	18	20	10	11

Solution:

Step 1 : **Hypotheses**

Null Hypothesis $H_0 : \mu_D = 0$

i.e., there is no significant increase in the mean sales after the training.

Alternative Hypothesis $H_1 : \mu_D > 0$

i.e., there is significant increase in the mean sales after the training. It is a one-sided alternative hypothesis.

Step 2 : **Data**

Sample size $n = 10$

Step 3 : **Level of significance**

$\alpha = 5\%$

Step 4 : **Test statistic**

Test statistic under the null hypothesis is

$$T = \frac{\overline{D}}{\frac{S}{\sqrt{n}}}$$

The sampling distribution of T under H_0 is t - distribution with (10–1) = 9 degrees of freedom.

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